

LAB 3: Inverse & "Pseudoinverse" of a Matrix

[See also James (1978) posted on Website]

⇒ Many linear systems reduce to problems of the form

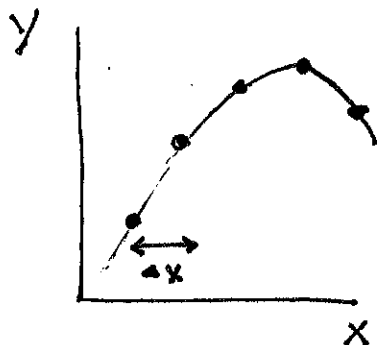
$$\begin{pmatrix} A_{11} & A_{12} & \cdots & \cdot \\ A_{21} & A_{22} & \cdots & \cdot \\ \vdots & \vdots & & \vdots \\ \cdot & \cdot & & \cdot \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ \cdot \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ \cdot \end{pmatrix}$$

$$\tilde{A} \tilde{X} = \tilde{Y}$$

Solve by inverting matrix $(\tilde{A}^{-1})(\tilde{A}) = \mathbb{1}$ "Identity Matrix"

$$\tilde{X} = \tilde{A}^{-1} \tilde{Y} \Rightarrow \text{so long as inverse exists}$$

⇒ Ex: Polynomial Interpolation



- Know (x_i, y_i) at discrete points
- Want approximation for arbitrary y
- Find polynomial that passes thru all points.

$$y(x) \approx a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

Factoids:

- Thm: For $n+1$ points $\exists$ one deg- n polynomial that passes thru all.
- Error in interpolation $\sim \Delta x^n$

⇒ For the case sketched above, a 4th order polynomial will interpolate all points.

$$\hookrightarrow \exists (a_0, a_1, a_2, a_3, a_4)$$

$$\text{S.T. } a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3 + a_4 x_i^4 = y_i \text{ for all } i$$

In matrix form

$$\begin{pmatrix} 1 & x_1 & x_1^2 & x_1^3 & x_1^4 \\ 1 & x_2 & x_2^2 & x_2^3 & x_2^4 \\ 1 & x_3 & x_3^2 & x_3^3 & x_3^4 \\ 1 & x_4 & x_4^2 & x_4^3 & x_4^4 \\ 1 & x_5 & x_5^2 & x_5^3 & x_5^4 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{pmatrix}$$

"Vandermonde Matrix" $\tilde{V}$

↪ invert $\tilde{V}$ with `numpy.linalg.inv`

$$\vec{a} = \tilde{V}^{-1} \vec{y}$$

< see vandermonde.ipynb >

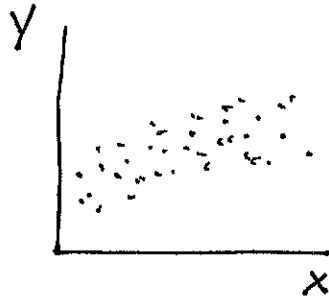
⇒ Now, all the above assumes $\tilde{A}$ is invertible.

Under what conditions is this true?

- $\det(\tilde{A}) \neq 0$ (rows linearly independent)
- $\tilde{A}$ is square Cramer $\tilde{A}^{-1} = \frac{C(\tilde{A})^T}{\det(\tilde{A})}$

⇒ What if $\tilde{A}$ is NOT square?

e.g. THE FITTING PROBLEM...



Model predicts $y = a_0 + a_1 x$

What is BEST LINEAR FIT?

IN MATRIX FORM

$$\begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \\ \vdots & \vdots \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{pmatrix}$$

WHAT IS THE INVERSE?

⇒ Generalized Approach : Singular Value Decomposition

$$\underline{\text{THM}} \quad \tilde{A} = \tilde{U} \tilde{S} \tilde{V}^+$$

$\nearrow$ $m \times n$ $\nearrow$ $m \times m$ $\nearrow$ $m \times n$ $\nwarrow$ $n \times n$ "unitary"
 "UNITARY" POSITIVE
 DIAGONAL

Unitary := Inverse = Conjugate Transpose

$$\tilde{U}^{-1} = (\tilde{U}^*)^T = \tilde{U}^+$$

and $\tilde{A}^{-1} = \tilde{V} \tilde{S}^{-1} \tilde{U}^+$

↳ Diagonals of $\tilde{S}$ called "singular values",
 if all $SV \neq 0$, $\exists \tilde{S}^{-1}$

⇒ Numerical issue: If any $SV \ll$ others, result will be a loss of precision

$$\text{"Condition Number"} = \frac{\text{Largest SV}}{\text{Smallest SV}}$$

if $CN \gg 1$, matrix is "ill-conditioned"

⇒ Strategy: set inverse of small $SV = 0$

Thus compute "pseudoinverse" / "generalized Inverse"

↳ Best Possible For Ill-Conditioned Matrix

<NOTEBOOK: Pseudoinverse.ipynb>

LAB (Due Today) - Slight variation on NOTEBOOK

HW01 (Due ~~Mon~~ 9/13) - Explicit calculation of
Pseudoinverse by SVD
AND QR methods.

`numpy.linalg.svd`

`numpy.linalg.qr`